

Reflective Journal

Ebuka

This lab helped me understand why exploratory data analysis is an important step before building a machine learning model. EDA involves reviewing summary statistics and visualizations to understand a dataset, identify missing information, and explore relationships. Using the Titanic dataset showed me how different ways of examining the same data can reveal different information. The workflow also showed how earlier machine learning tools fit into a larger process. Before selecting or training a model, the data needs to be understood and prepared. EDA can identify missing values, categorical columns, and other issues that must be addressed in later stages. Each step builds on the previous one, making the order of the workflow important.

Sadia

One thing I found challenging during this lab was getting some of the code cells to run correctly. Some of the code was placed on one line, which caused syntax errors. I had to go through the cells, separate the commands correctly, and run them again until the visualizations worked. This helped me understand that checking the code and outputs is an important part of the workflow and not just something to do at the end.

I also learned more from the visualizations than I expected. Looking at survival by gender and passenger class made the patterns much easier to understand than just looking at numbers. I could clearly see that female and first-class passengers had better survival rates. This helped me understand why EDA is useful before building a model because visualizations can reveal relationships that summary statistics alone may not show.

Sam

Reflection Journal: Foundations of Machine Learning Data Preparation

Introduction and Initial Exploration

Embarking on the data preparation phase of a machine learning project often feels like stepping into a meticulous workshop where raw, unrefined materials must be shaped before any true engineering can begin. Reviewing the steps taken to process the Titanic dataset—specifically handling missing values, encoding categorical text, and executing feature scaling—highlights a foundational truth of data science: algorithms are mathematical engines. They do not possess intuition, context, or the ability to gloss over

messy inputs. Without proper preparation, a model will inevitably struggle to learn correctly or produce heavily biased results.

The initial inspection of the dataset immediately revealed common real-world data flaws. In raw datasets, missing values are an expected obstacle. If left unaddressed, models like logistic regression or support vector machines will either crash due to algebraic incompatibility or force the execution to discard valuable rows entirely. Rather than blindly dropping data, this lab utilized median imputation to fill in missing ages. This choice carried distinct structural advantages. Because human age distributions can feature skewness or rare outliers, the median provides a robust measure of central tendency that resists distortion. By imputing missing values with the median, we preserved dataset integrity and maximized our sample size without introducing artificial skew.

Structural Transformations and Scaling

Moving past missing values, the next hurdle was addressing structural incompatibility with categorical data. Machine learning models cannot natively process text labels such as "male," "female," or port designations like "C," "Q," and "S." A common novice misstep is mapping these text categories to arbitrary integers (such as assigning 1, 2, and 3). However, doing so introduces a false mathematical hierarchy, tricking the algorithm into assuming that category 3 is three times "greater" than category 1, or that a linear geometric distance exists between them. One-hot encoding solved this by expanding categories into binary indicator columns. This preserved categorical distinctions cleanly as zeros and ones, allowing the model to evaluate the true presence or absence of a trait without assuming false numerical rankings.

Finally, feature scaling via `StandardScaler` addressed the distortion caused by disparate measurement scales. In the raw Titanic dataset, features exist in entirely different universes: age might range from 0 to 80, while ticket fare can span from 0 to over 500. Distance-based algorithms and gradient descent optimizers naturally assign disproportionate weight to larger numerical magnitudes, meaning an unscaled Fare feature would completely overpower Age during optimization. Standardizing numeric columns to a mean of zero and a standard deviation of one leveled the playing field, ensuring that every feature contributed equitably to the model's convergence and learning trajectory.

Final Takeaway

Data preparation is a critical part of the machine learning pipeline because the quality of the input data directly affects model performance. Cleaning, encoding, and scaling the data helps the model learn more accurately and reliably. Ultimately, this exercise reinforced that building a high-performing model is less about tuning complex algorithms and more about disciplined data hygiene. Transforming raw, messy real-world data into a pristine, mathematically sound matrix is where the true engineering work happens.